Netflix Content Trends & Genre Evolution

Manpreet Kaur – 33647884
Vlad Petrariu – 76370956

Project Links

Demo: https://www.youtube.com/watch?v=6il7pzQwsB4
GitHub Repository: https://github.com/VladPetrariu/netflix-content-analytics.git

1. Research Question

This project analyzes how Netflix's content library has evolved over time in terms of:

- Genre popularity and emerging categories
- Content type balance (movies vs. TV shows)
- Geographic diversity of production countries
- Duration patterns for movies and TV show seasons

2. Dataset & Source

The dataset used is the **Netflix Movies and TV Shows dataset** from Kaggle:
https://www.kaggle.com/datasets/shivamb/netflix-shows

It contains ~6,200 titles with attributes such as:

- Title, type (movie/TV show)
- Director, cast, country
- Date added, release year
- Rating, duration
- Genres

This dataset enables analysis of content growth, genre trends, and geographic contributions.

3. Applied Task

This project performs **information extraction** by grouping and aggregating data to identify patterns such as:

- Genre trends over time

- Country contributions
- Growth of movies vs TV shows

This aligns with the required applied analytics task.

4. Tools Used

- **Python (pandas, matplotlib, seaborn)** → data cleaning & analysis
- **SQL (SQLite)** → structured storage and querying
- **Tableau** → dashboards and visualization
- **Excel** → supplementary validation

5. Pipeline Overview

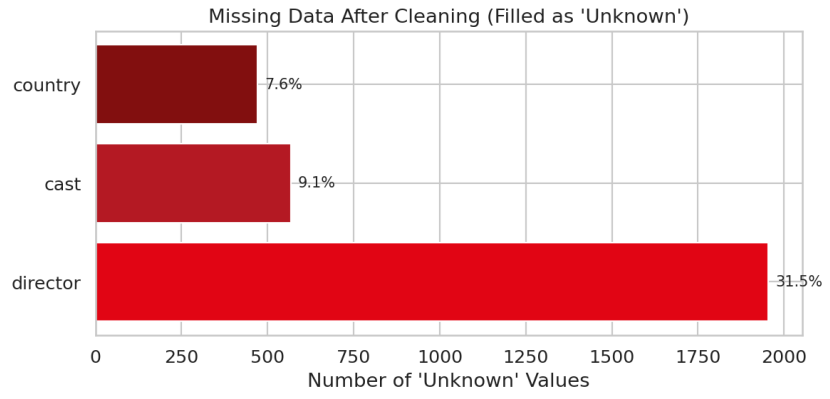
5.1 Data Acquisition

- Dataset downloaded from Kaggle
- Raw data stored in `data/raw/`

5.2 ETL & Data Cleaning

- Filled missing values:
  - Director, cast, country → `"Unknown"`
- Removed rows missing critical fields:
  - `date_added`, `rating`

Data Quality Check (Missingness)
Figure 1: Missing values heatmap



- Converted `date_added` → datetime
- Split duration into numeric + unit
- Created:
    - `year_added`, `month_added`
    - `primary_country`, `primary_genre`

Final dataset: **6,214 rows, 18 columns**

5.3 Data Storage

- Stored in SQLite database (`netflix.db`)
- Indexed columns for efficient querying

5.4 Exploratory Data Analysis (EDA)

Performed:

- Distribution analysis
- Trend analysis
- Relationship analysis

6. Results

6.1 Content Growth Timeline

Netflix experienced rapid growth from **2015–2019**, peaking in **2019**.

Figure 2: Titles per Year

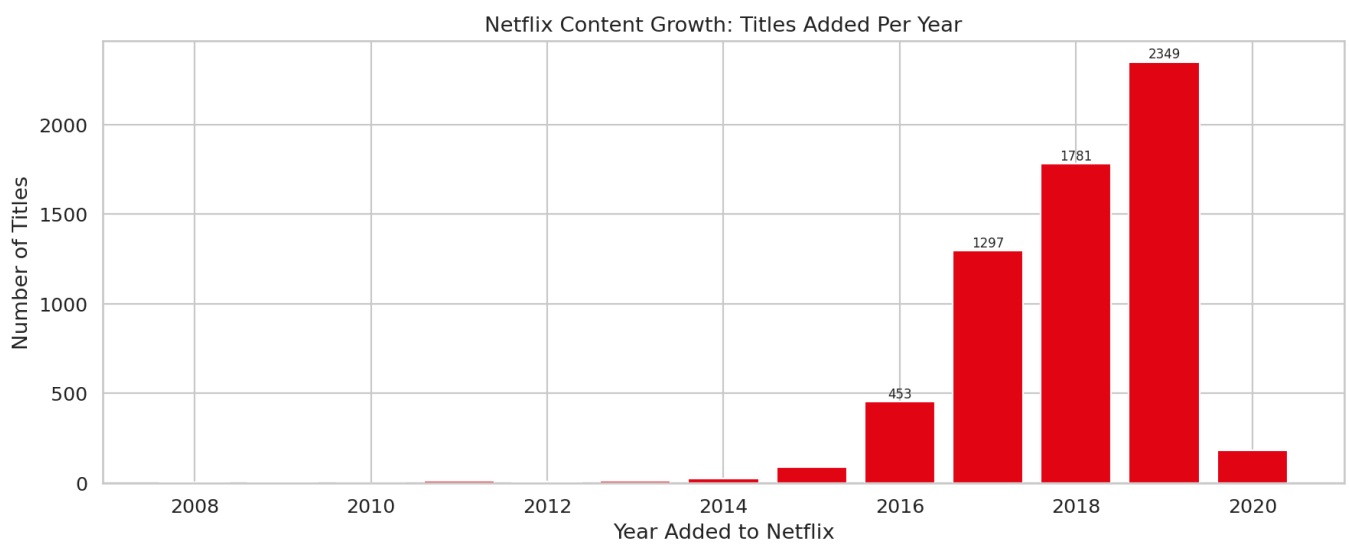
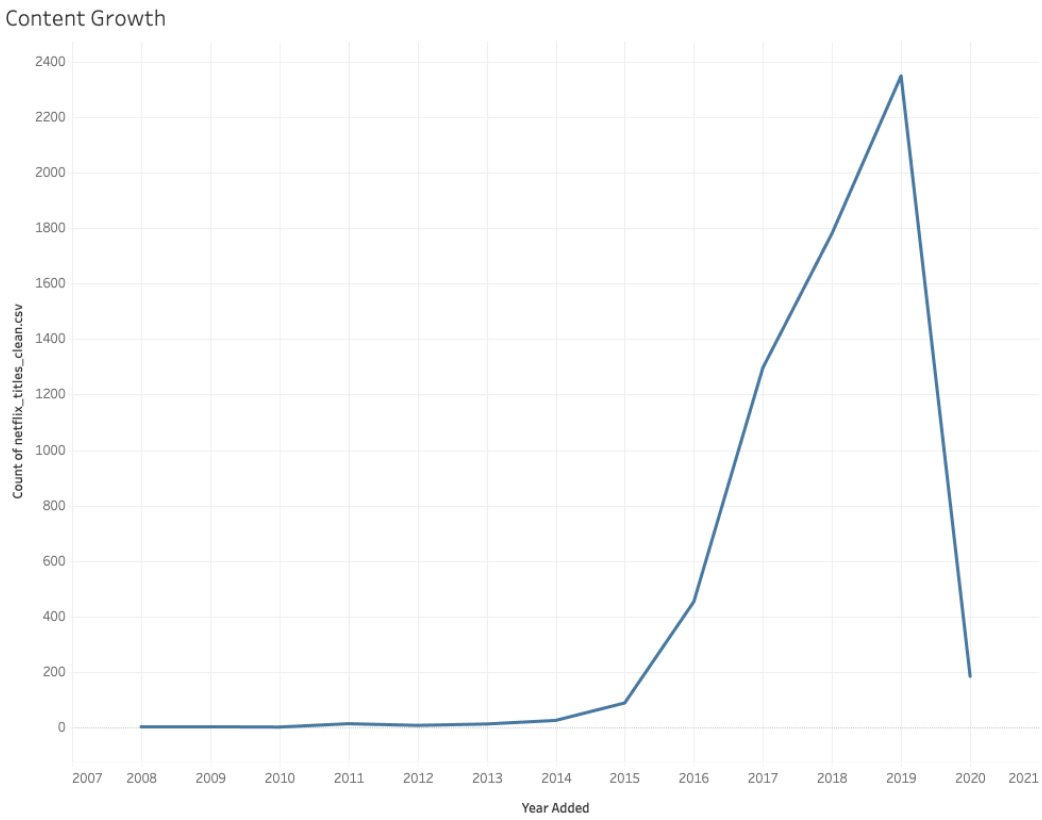


Figure 3: Tableau Dashboard – Content Growth



6.2 Genre Analysis

Figure 4: Genre Frequency (Overall)

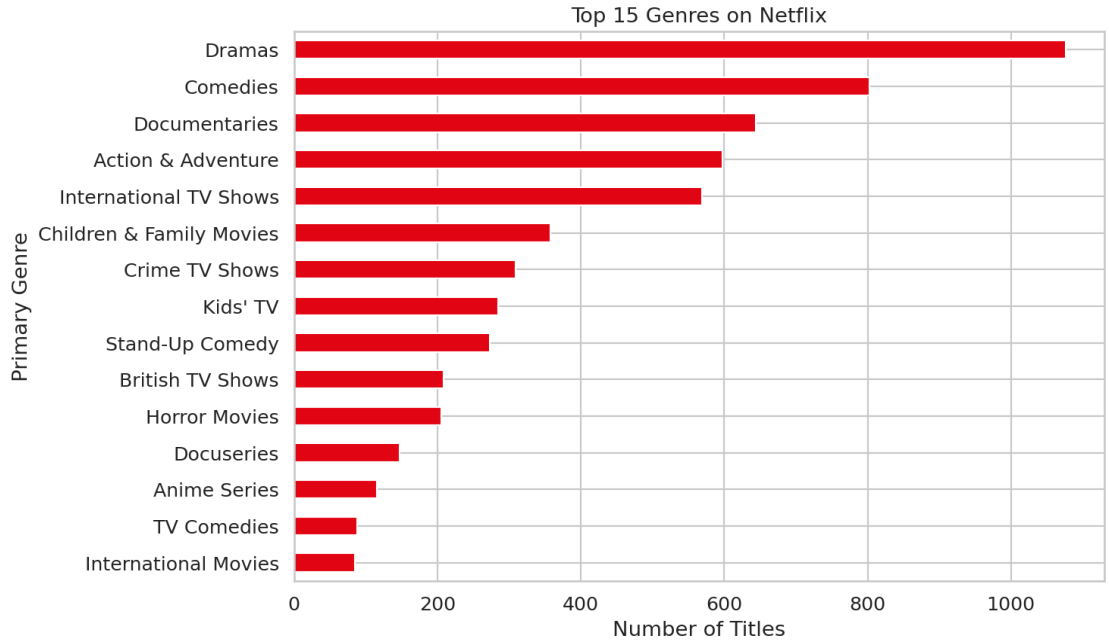


Figure 5: Genre Trends Over Time

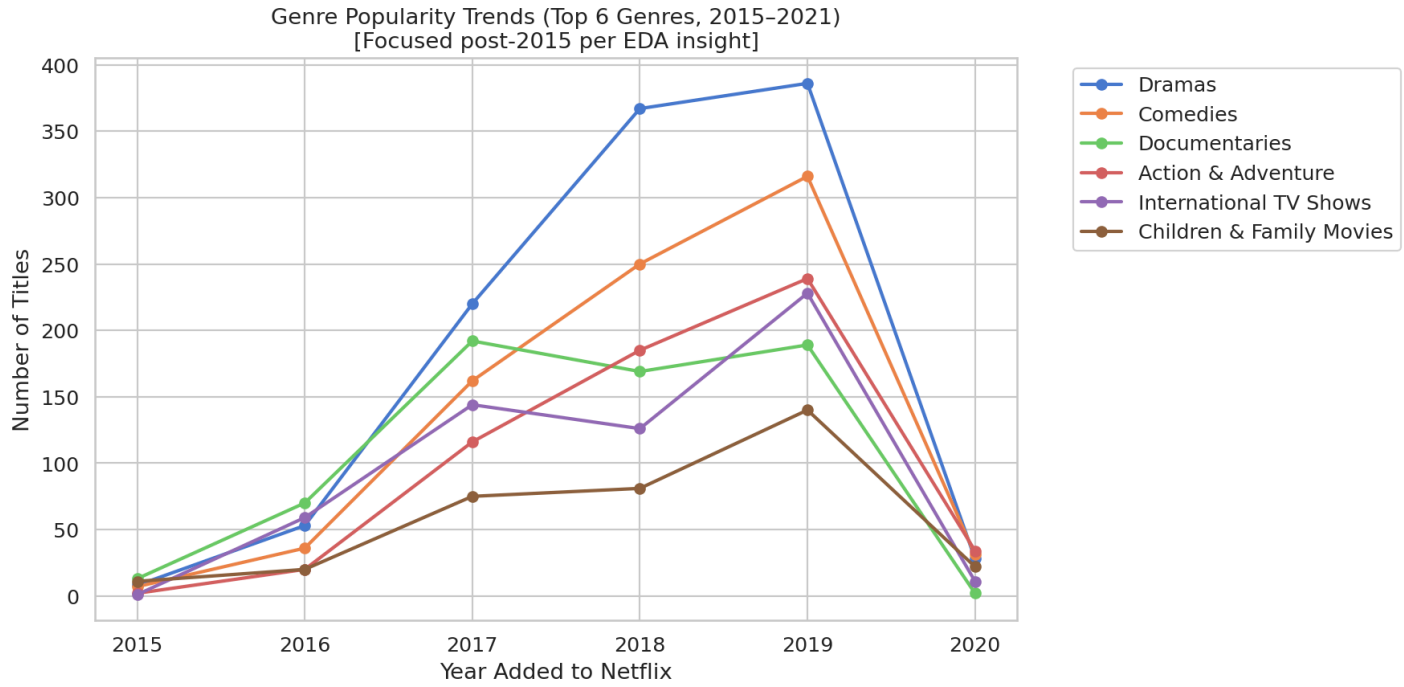


Figure 6: Genre vs Year Heatmap

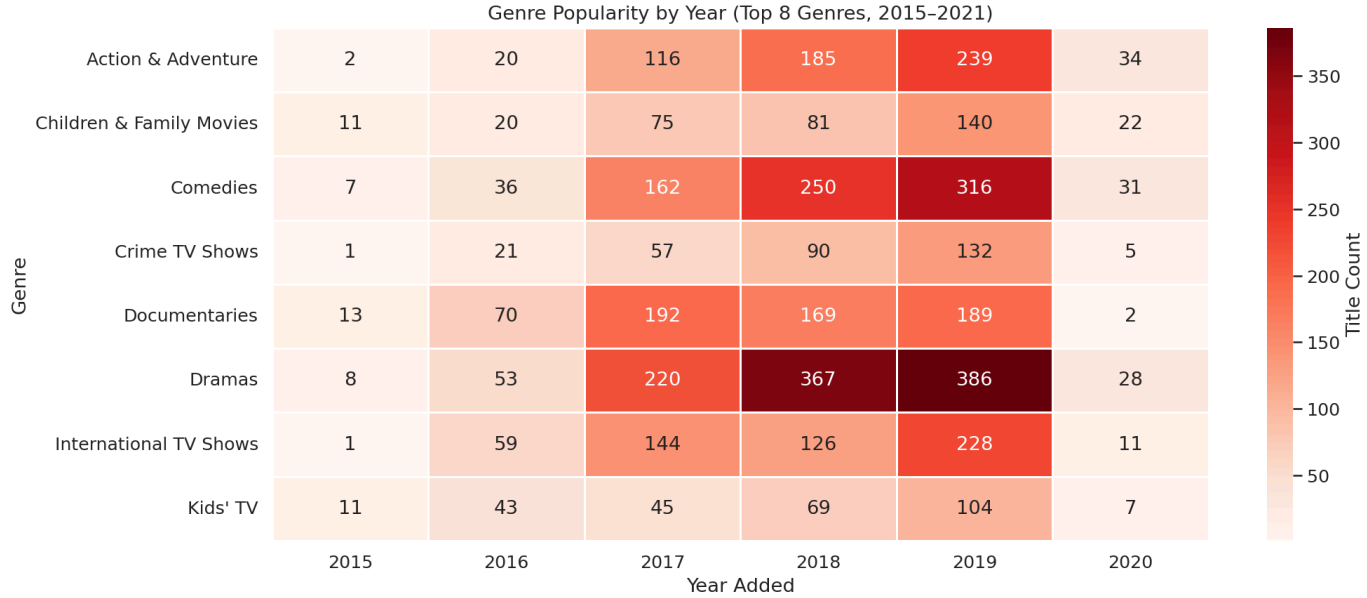
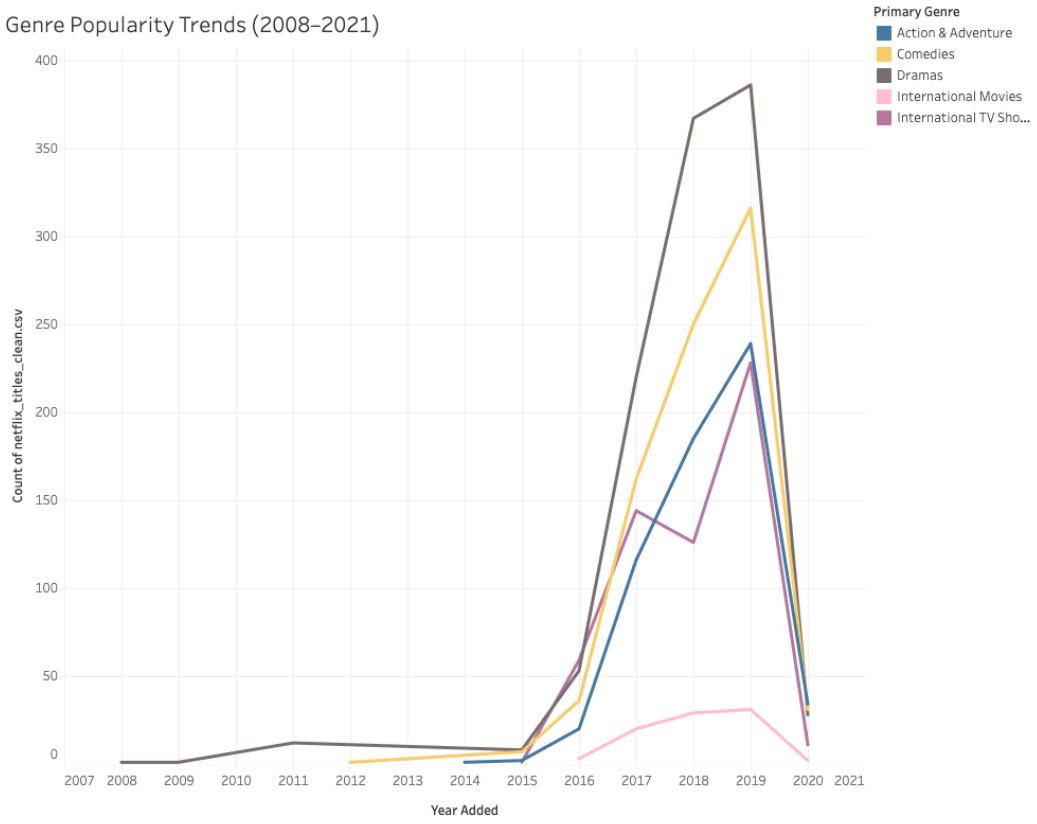


Figure 7: Tableau – Genre Trends



Insights:

- International Movies and Dramas dominate
- Documentaries & Stand-Up Comedy increased after 2016

6.3 Country Contributions

Figure 8: Country Frequency

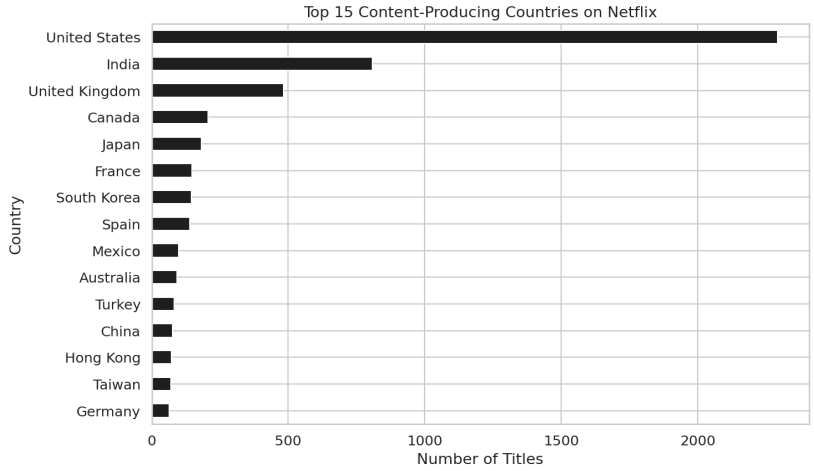


Figure 9: Country Contribution Trends

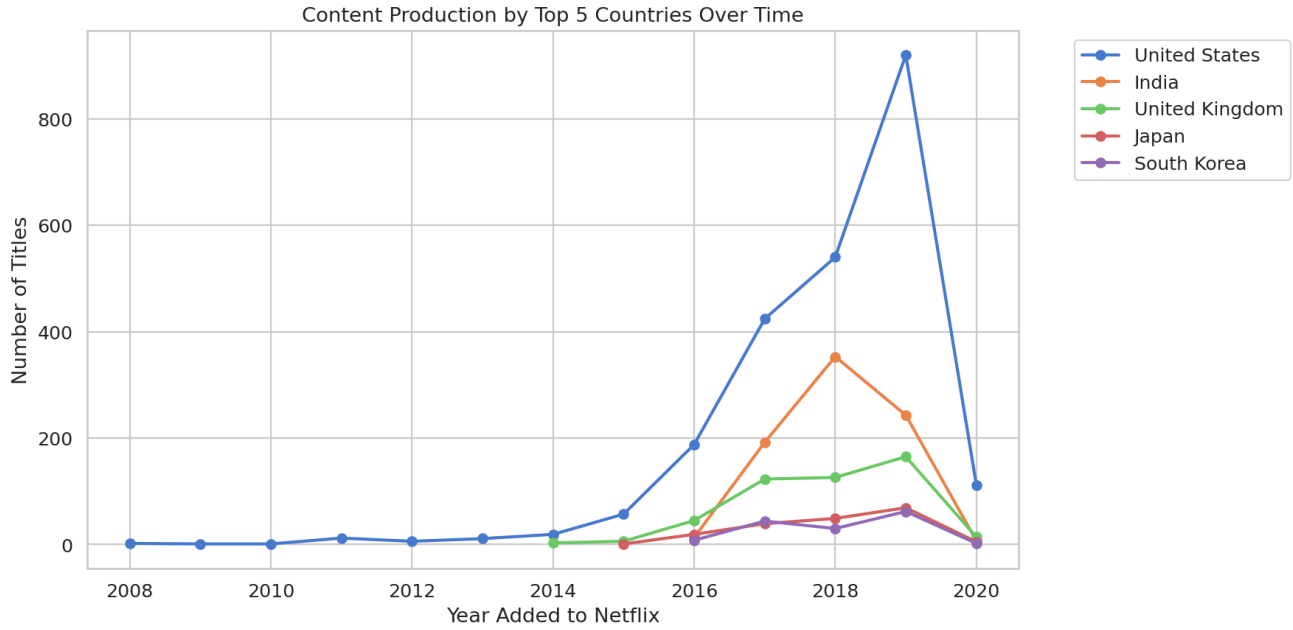


Figure 10: Country vs Content Type

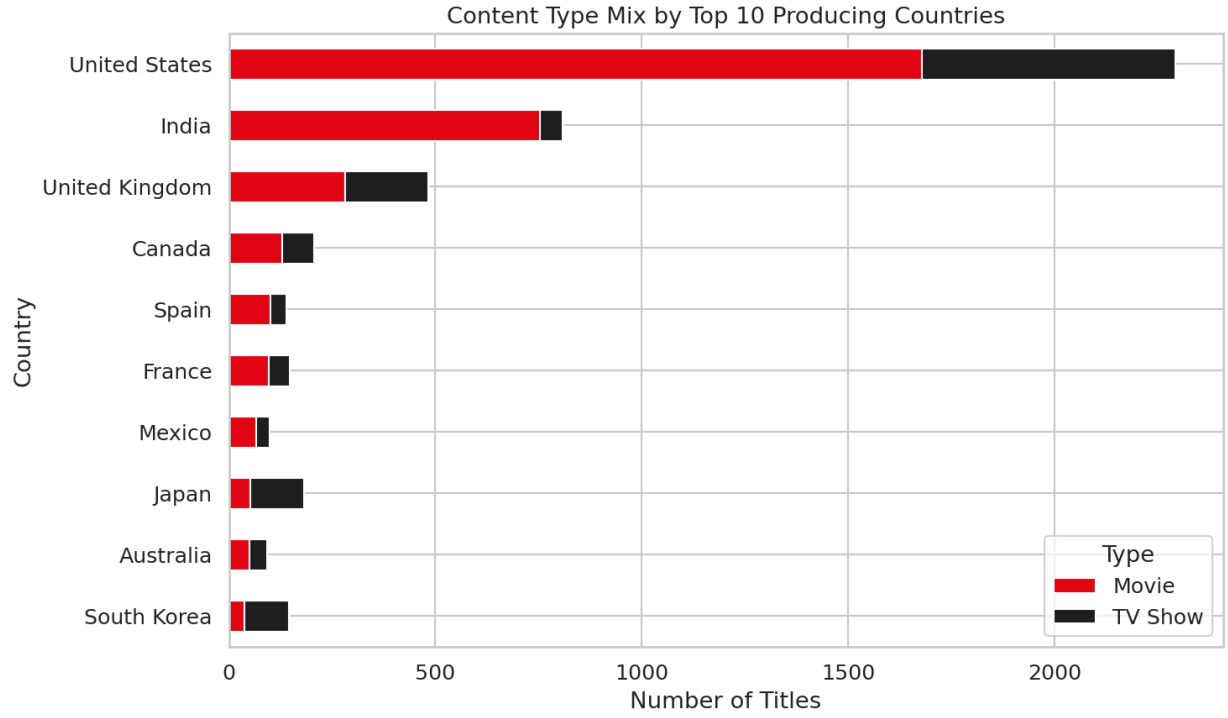
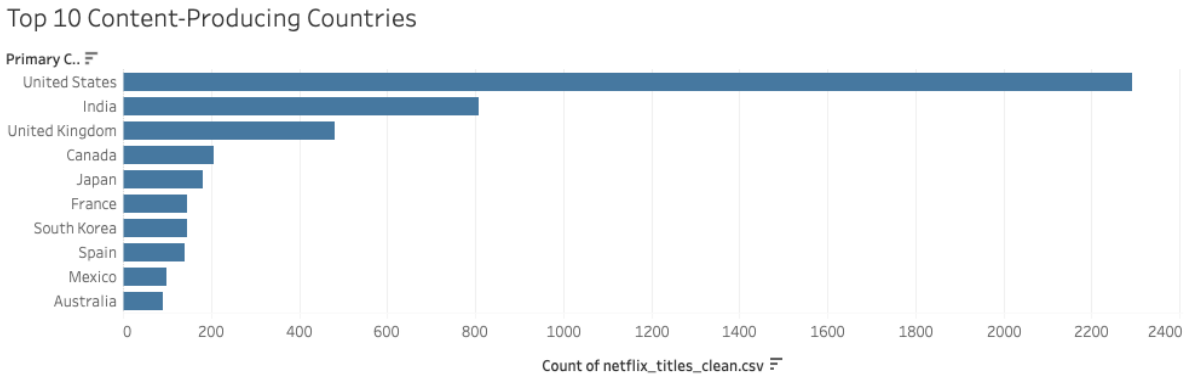


Figure 11: Tableau – Top Countries



Insights:

- USA dominates (~2300 titles)
- India is second (~800 titles)
- Growth seen in South Korea & Japan

6.4 Movies vs TV Shows

Figure 12: Movies vs TV Shows Over Time

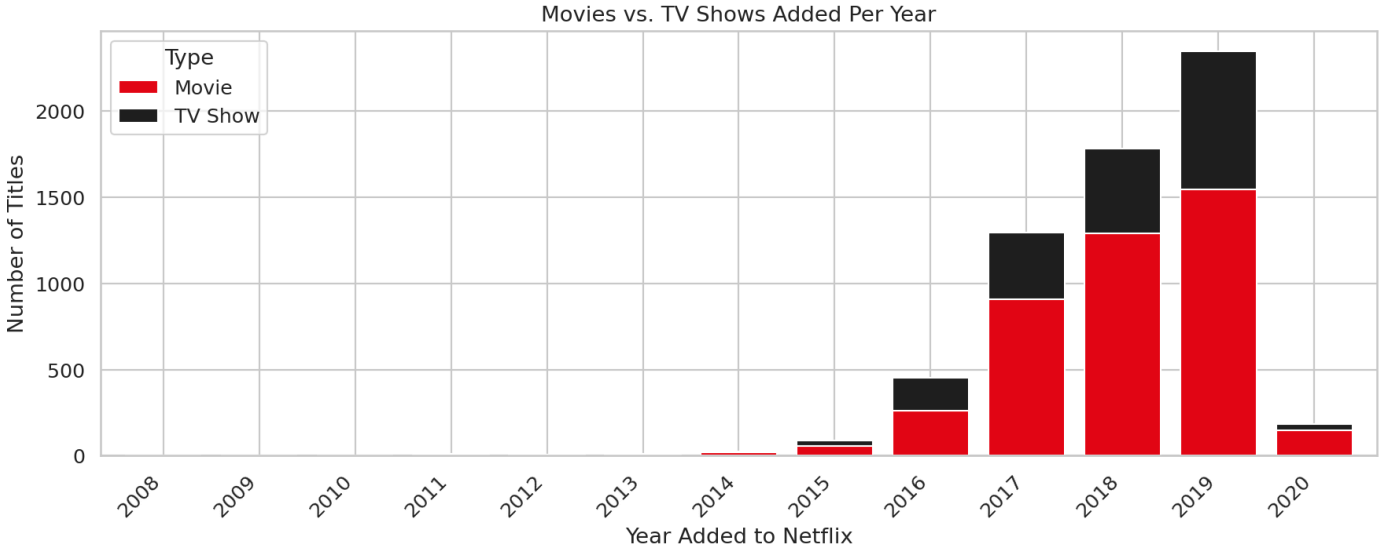


Figure 13: Overall Movies vs TV Shows

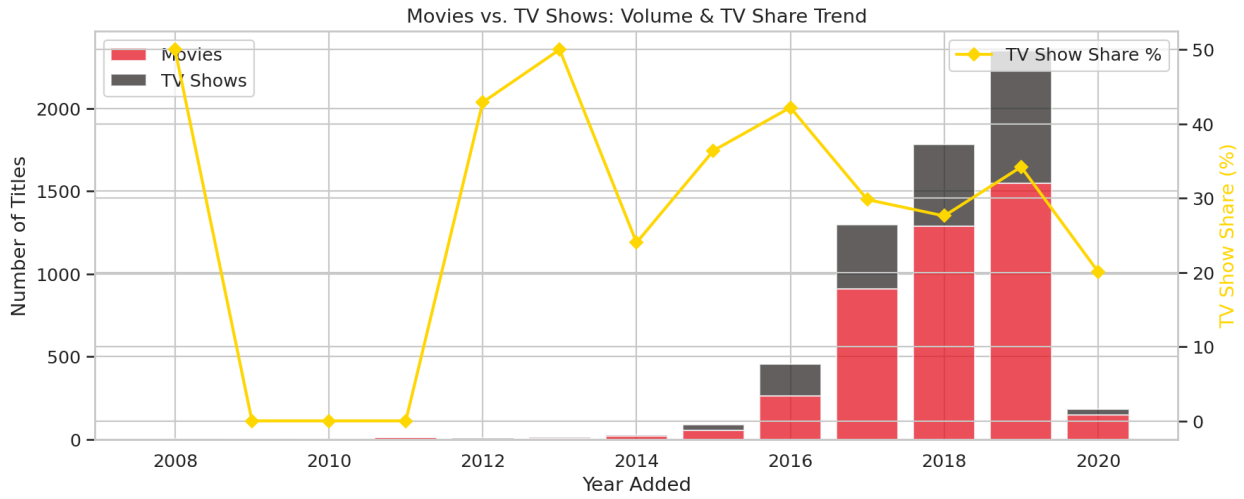
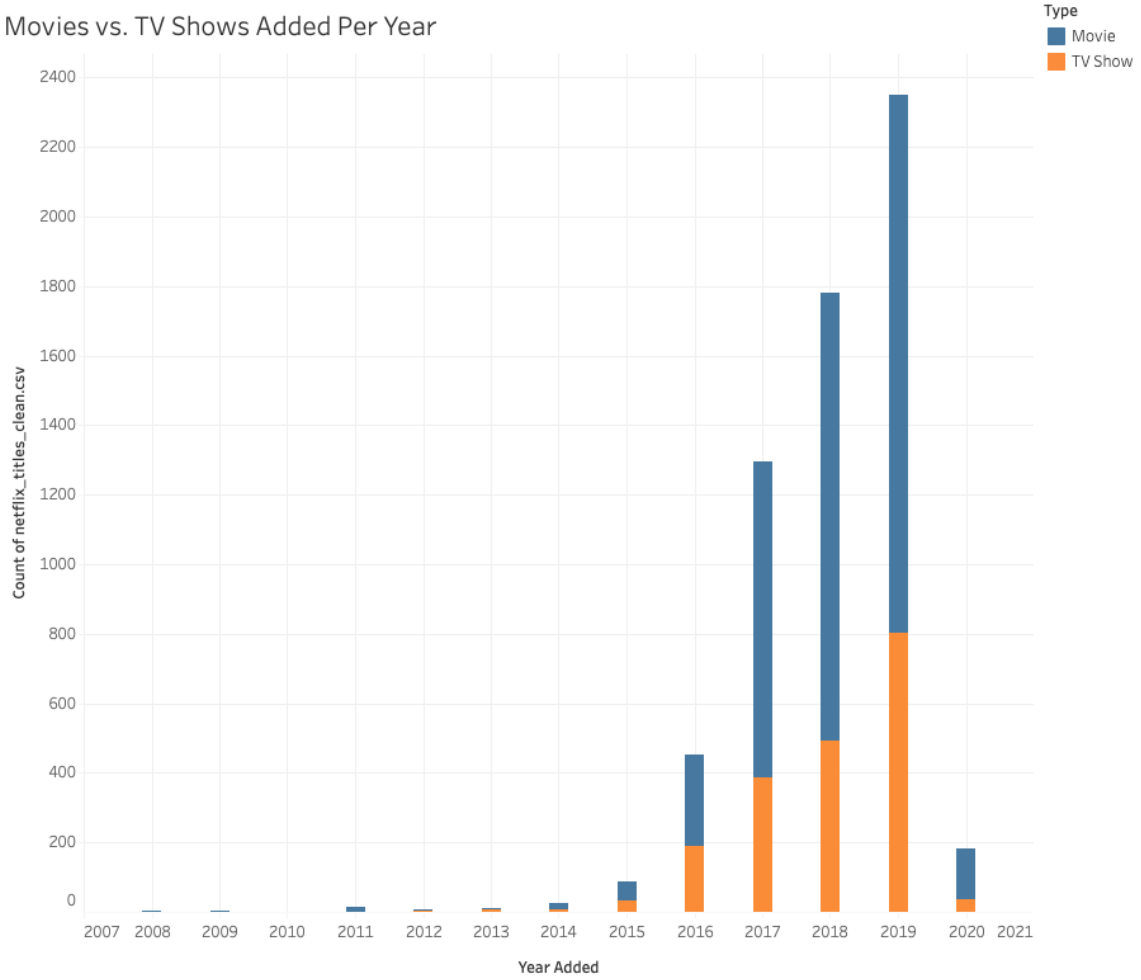


Figure 14: Tableau – Movies vs TV

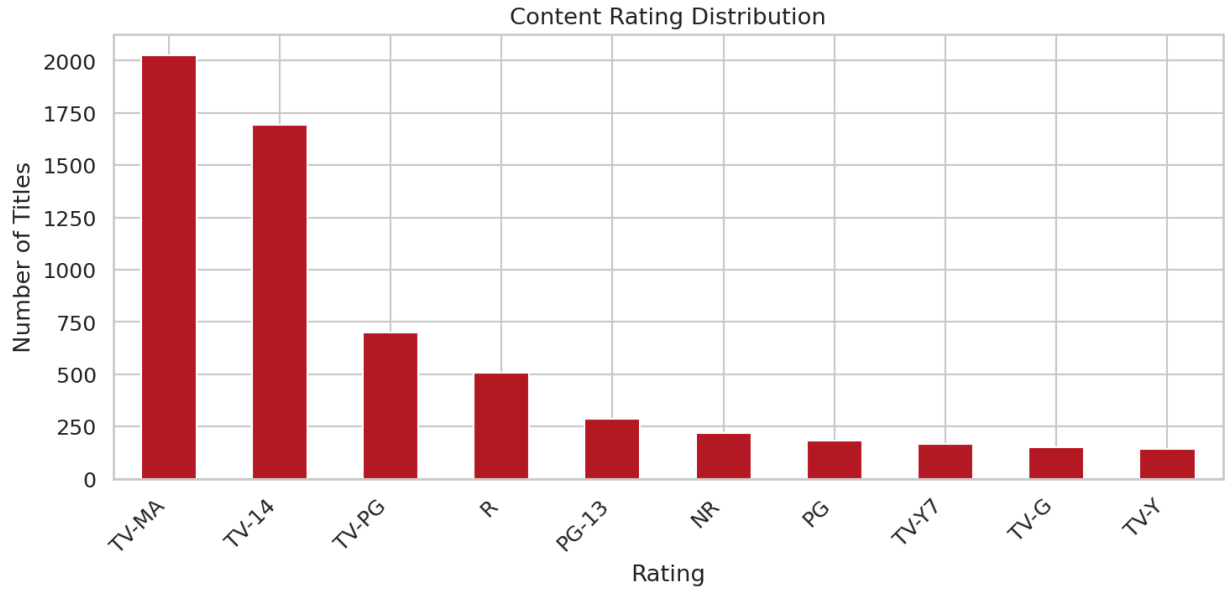


Insights:

- Movies dominate overall
- TV shows are increasing significantly

6.5 Content Rating Distribution

Figure 15: Content Ratings



Insight:

- TV-MA dominates → more mature content focus

6.6 Duration Analysis

Figure 16: Movie Duration Distribution

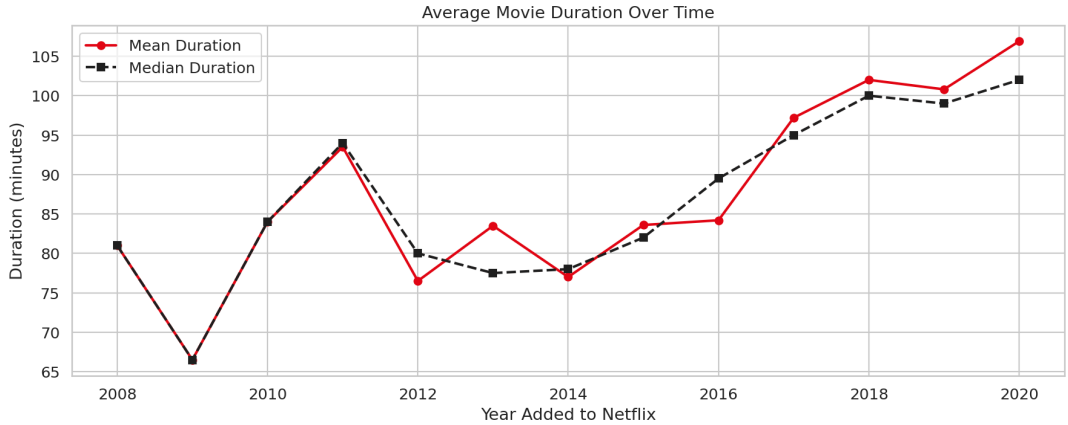
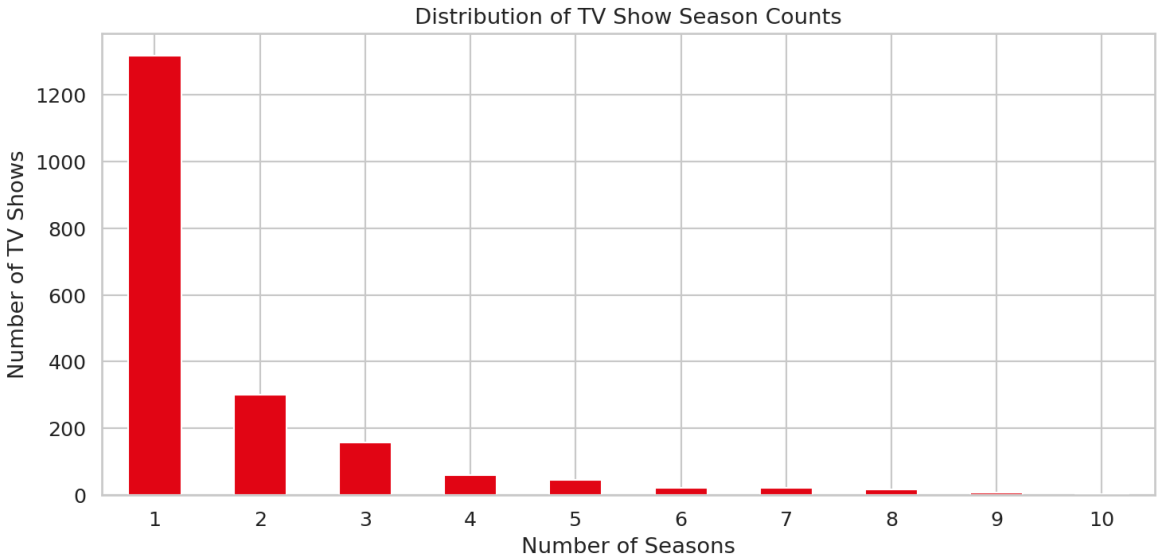


Figure 17: TV Show Season Distribution



Insights:

- Movies stable (~90–100 min)
- **67.4% of TV shows have 1 season**

7. Key Insights

1. Content additions peaked in **2019**
2. **Dramas & International Movies dominate**
3. **India became second-largest contributor**
4. Shift toward **TV shows**
5. Rise of **Documentaries & Stand-Up Comedy**
6. Most TV shows are **single-season**
7. Content increasingly targets **mature audiences**

8. Limitations

- Dataset is a **snapshot (mid-2021)**
- No data on removed content
- Multi-genre simplified
- Missing values may bias results
- No viewership data
- Cannot infer causation

9. Reproducibility

To reproduce:

1. Download dataset from Kaggle
2. Run Python cleaning scripts
3. Load data into SQLite
4. Run SQL queries / Tableau dashboards

Folder Structure

```
data/
 raw/
 clean/
scripts/
database/
visualizations/
dashboards/
README.md
```

10. References

- Kaggle Dataset: Netflix Movies and TV Shows
- Tools: Python, SQLite, Tableau, Excel